

VAPOUR-LIQUID EQUILIBRIA IN ASSOCIATING SOLUTIONS

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Abstract—Isobaric vapour-liquid equilibria data were measured for the binary mixtures of acetic acid-propionic acid and formic acid-propionic acid at 760 mm Hg. The data were tested against a new thermodynamic model which extends the one proposed by Marek in that it is suitable for the case of binary mixtures containing two associating species and takes into account hetero-dimerization in the vapour phase. The proposed model converts into that of Christian for the particular case of an ideal liquid solution. The model fits well the system acetic acid-propionic acid with regard to the thermodynamic consistency of the overall activity coefficients, but does not fit the second mixture so well, probably because higher degrees of polymerization have to be included in it.

INTRODUCTION

It has definitely been established that in binary mixtures containing carboxylic acids, pure or dissolved in hydrocarbons and similar non polar solvents, the acid monomer undergoes partial dimerization and even higher polymerization [1-5]. This association is attributed to the formation of hydrogen bonds and occurs in both the vapour and liquid phase. If the mixture contains two carboxylic acids, the association phenomena become more complicated since hetero-dimerization between the two acids can also take place. The association phenomena were long ago recognized to be significant for the correct interpretation of experimental data and their inclusion in vapour-liquid equilibria relations has helped in explaining the anomalous behaviour of fatty acids in the vapour and liquid phase.

Marek and Standard [6, 7] were among the first to propose equilibrium relationships that included the effect of association for systems that contained one component which was partly dimerized in both phases while the other species did not undergo association. Application of these relationships to binary mixtures of acetic acid with water, benzene and *p*-xylene yielded activity coefficients that were thermodynamically consistent. Wisniak and Tamir [8] recently applied Marek's relationships to the system carbon tetrachloride-acetic acid and showed that they produced consistent data for the activity coefficients. Tetsuo and Yoshida [9] investigated the vapour-liquid equilibria of water-lower fatty acid systems and in their calculation of the activity coefficients accounted for the formation of trimer and dimers from the acids. A similar approach was adopted by Sebastiani and Lacquaniti for the system acetic acid-water [10].

The experimental work on two associating acids is

meager and has been performed mainly by Christian [11], who obtained isothermal data at 20°C for the system acetic acid-propionic acid. Assuming ideal behaviour of the liquid phase, he developed a model which accounted for dimerization and hetero-dimerization between the acids and reasonably fitted his experimental data. In a later paper, Affsprung *et al.* [12] studied more thoroughly the hetero-association of acetic and propionic acids in CCl₄. It was found that formation of a heterodimer was in preference to that of homodimers, a fact which was also in agreement with the observations of Hausen and Christian [12] on their vapour-liquid equilibria study of the systems acetic acid-trifluoroacetic acid, propionic acid-pentafluoropropionic acid and others.

This work was motivated by the relatively scarce data on the behaviour of binary mixtures containing species which tend to associate and to form homo and heterodimers. Isobaric vapour-liquid equilibria data were measured for the binary mixtures of formic acid-propionic acid and acetic acid-propionic acid at 760 mm Hg. The data were tested against a new thermodynamic model which extends the one proposed by Marek [6, 7] in that it is suitable for the case of binary mixtures containing two associating species and takes into account hetero-dimerization in the vapour phase. The proposed model converts into that of Christian's [11] for the particular case of an ideal liquid solution.

VAPOUR-LIQUID EQUILIBRIA RELATIONSHIPS

Assume *A* and *B* to be two species undergoing association. Their vapour or liquid mixture contains the monomers *A*₁ and *B*₁, the dimers *A*₂ and *B*₂ and the heterodimer *AB*. In a mixture containing *n*_{A1}, *n*_{A2}, *n*_{B1}, *n*_{B2}, and *n*_{AB} moles, the true mole fraction *η*_A, of the monomer,

for example, is given by

$$\eta_{A_1} = \frac{n_{A_1}}{n_{A_1} + n_{A_2} + n_{B_1} + n_{B_2} + n_{AB}} \quad (1)$$

The stoichiometric mole fraction y_A of species A which includes the monomer and the dimer reads

$$y_A = \frac{n_A}{n_A + n_B} = \frac{n_{A_1} + 2n_{A_2} + n_{AB}}{(n_{A_1} + 2n_{A_2} + n_{AB}) + (n_{B_1} + 2n_{B_2} + n_{AB})} \quad (2)$$

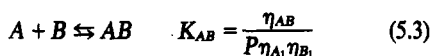
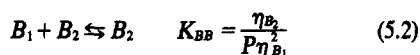
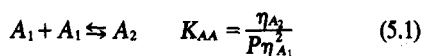
In terms of the true mole fractions, it is given by

$$y_A = \frac{\eta_{A_1} + 2\eta_{A_2} + \eta_{AB}}{1 + \eta_{A_2} + \eta_{B_2} + \eta_{AB}} \quad (3)$$

and similarly

$$y_B = \frac{\eta_{B_1} + 2\eta_{B_2} + \eta_{AB}}{1 + \eta_{A_2} + \eta_{AB}} \quad (4)$$

It is assumed that the following reversible reactions take place with the indicated equilibrium constants for association in the vapour phase:



Substituting Eqs. (5) in Eqs. (3) and (4) yields for the unknowns η_{A_1} and η_{B_1}

$$K_{AA}P(1 + y_B)\eta_{A_1}^2 + \eta_{A_1} - y_A(1 + K_{BB}P\eta_{B_1}^2) + K_{AB}P\eta_{A_1}\eta_{B_1}y_B = 0 \quad (6)$$

$$K_{BB}P(1 + y_A)\eta_{B_1}^2 + \eta_{B_1} - y_B(1 + K_{AA}P\eta_{A_1}^2) + K_{AB}P\eta_{A_1}\eta_{B_1}y_A = 0 \quad (7)$$

Assuming ideal gas behaviour in the vapour phase, the equilibrium condition for the monomer is given

$$P\eta_{A_1} = \gamma_{A_1}\xi_{A_1}P_{A_1}^\circ \quad (8)$$

where γ_{A_1} is the activity coefficient for the monomer in the mixture and η_{A_1} , ξ_{A_1} are its true mole fraction in the vapor and liquid phase respectively. η_{A_1} (and η_{B_1}) may be calculated from the simultaneous solution of Eqs. (6) and

(7) if the equilibrium constants in the vapour phase are available. ξ_{A_1} may also be expressed in terms of liquid equilibrium constants as spelled out by Marek[6, 7]; however, values of these constants are usually not available. The vapour pressure of the pure substance, $P_{A_1}^\circ$, is the sum of the partial pressures of the monomer $P_{A_1}^\circ$, and dimer $P_{A_2}^\circ$, so that

$$P_A^\circ = P_{A_1}^\circ + P_{A_2}^\circ = P_{A_1}^\circ + K_{AA}(P_{A_1}^\circ)^2 \quad (9)$$

One can obtain from Eq. (9) that

$$P_{A_1}^\circ = \frac{-1 + \sqrt{1 + 4K_{AA}P_A^\circ}}{2K_{AA}} \quad (10)$$

Equation (8) may be written as

$$Py_A Z_A = \gamma_{A_1} X_A P_{A_1}^\circ \quad (11)$$

where

$$Z_A = \frac{\eta_{A_1}}{y_A} \quad \text{and} \quad \gamma_{A_1} = \gamma_{A_1} \left(\frac{\xi_{A_1}}{X_A} \right) \quad (12)$$

The factor Z_A expresses the influence of the vapour phase association of A and as mentioned before, it may be evaluated from the knowledge of the association constants in Eqs. (6) and (7). γ_A is an overall activity coefficient which is obtained from experimental measurements. It should be noted that it is impossible to obtain the values of γ_{A_1} in Eq. (12) or (8) separately.

Similarly for species B ,

$$P\eta_{B_1} = \gamma_{B_1}\xi_{B_1}P_{B_1}^\circ \quad (13)$$

or alternatively

$$Py_B Z_B = \gamma_B X_B P_{B_1}^\circ \quad (14)$$

where

$$Z_B = \frac{\eta_{B_1}}{y_B} \quad \text{and} \quad \gamma_B = \gamma_{B_1} \left(\frac{\xi_{B_1}}{X_B} \right) \quad (15)$$

$P_{B_1}^\circ$ is given by an equation similar to (10). Substitution of η_{A_1} and η_{B_1} from Eqs. (11) and (13) into Eqs. (6) and (7), makes it possible to calculate γ_A and γ_B by the simultaneous solution of the general equations

$$\begin{aligned} &K_{AA}(1 + y_B)(\gamma_A X_A P_{A_1}^\circ)^2 + \gamma_A X_A P_{A_1}^\circ \\ &\quad - y_A [P + K_{BB}(\gamma_B X_B P_{B_1}^\circ)^2] \\ &\quad + K_{AB}(\gamma_A X_A P_{A_1}^\circ)(\gamma_B X_B P_{B_1}^\circ)y_B = 0 \end{aligned} \quad (16)$$

$$\begin{aligned}
& K_{BB}(1+y_A)(\gamma_B X_B P_{B_1}^\circ)^2 + \gamma_B X_B P_{B_1}^\circ \\
& - y_B [P + K_{AA}(\gamma_A X_A P_{A_1}^\circ)^2] \\
& + K_{AB}(\gamma_A X_A P_{A_1}^\circ)(\gamma_B X_B P_{B_1}^\circ) y_A = 0
\end{aligned} \quad (17)$$

The solution can be aided by an iterative technique that utilizes the Newton-Raphson procedure[15] to ensure fast convergence.

For the particular case of an ideal liquid solution where $\gamma_A = \gamma_B = 1$, Eqs. (16) and (17) reduce to

$$\begin{aligned}
y_A = & \frac{X_A P_{A_1}^\circ + 2K_{AA}(X_A P_{A_1}^\circ)^2 + K_{AB}(X_A P_{A_1}^\circ)(X_B P_{B_1}^\circ)}{P + K_{AA}(X_A P_{A_1}^\circ)^2 + K_{BB}(X_B P_{B_1}^\circ)^2 + K_{AB}(X_A P_{A_1}^\circ)(X_B P_{B_1}^\circ)}
\end{aligned} \quad (18)$$

$$\begin{aligned}
y_B = & \frac{X_B P_{B_1}^\circ + 2K_{BB}(X_B P_{B_1}^\circ)^2 + K_{AB}(X_A P_{A_1}^\circ)(X_B P_{B_1}^\circ)}{P + K_{AA}(X_A P_{A_1}^\circ)^2 + K_{BB}(X_B P_{B_1}^\circ)^2 + K_{AB}(X_A P_{A_1}^\circ)(X_B P_{B_1}^\circ)}
\end{aligned} \quad (19)$$

which are similar to those obtained by Christian[11]. Equations (18) and (19) yield the following quadratic equation for X_A which can be solved for given values of P and T

$$\begin{aligned}
& [K_{AB}P_{A_1}^\circ P_{B_1}^\circ - K_{AA}(P_{A_1}^\circ)^2 - K_{BB}(P_{B_1}^\circ)^2]X_A^2 \\
& - [K_{AB}P_{A_1}^\circ P_{B_1}^\circ - 2K_{BB}(P_{B_1}^\circ)^2 + P_{A_1}^\circ - P_{B_1}^\circ]X_A \\
& + [P - K_{BB}(P_{B_1}^\circ)^2 - P_{B_1}^\circ] = 0
\end{aligned} \quad (20)$$

EXPERIMENTAL

Purity of materials

Analytical grade reagents were purchased from Merck and Frutarom and employed without further purification after gas chromatographic analysis failed to show any significant impurities. The propionic acid was at least 99.5% pure. The physical properties of the pure compounds appear in Table 1.

Table 1. Physical constants of compounds at 25°C

Compound	Density	Refractive index
Formic acid	1.2136	1.3690
	1.2139 [14]	1.3693 [14]
Acetic acid	1.0435	1.3696
	1.0437 [14]	1.3698 [14]
Propionic acid	0.9879	1.3837
	0.9880 [14]	1.3843 [14]

Apparatus and procedure

An all-glass modified Dvořák and Boublik recirculation still[13] was used in the equilibrium determinations. The original apparatus was further modified after pressure equalization problems were encountered when working at temperatures above 90°C. Recirculation of the condensed vapour phase was substantially improved by providing a connection between the entrance to the vapour condenser and the condensed vapour reservoir. A vacuum system controlled by a cartesian manostat connected the vapour condenser with a Swietosławski ebulliometer and allowed total pressure regulation. The total pressure of the system was determined from the boiling temperature of distilled water in the ebulliometer. Temperatures were measured with a Hewlett-Packard Quartz thermometer, Model 2801A, with an accuracy of $\pm 0.005^\circ\text{C}$, so that the accuracy of the pressure measurements was better than 0.1 mm Hg. Several 1 ml samples were taken during each run until no significant changes in the refractive index and temperature were detected. The time required for reaching steady state varied between 2 hr for low temperature determinations, to 4 hr for high temperature levels.

Analyses of the samples were made at 25°C with a Bausch & Lomb Abbe-3L precision refractometer which gave direct readings to four significant figures and estimated values reproducibly to the nearest 0.00025. The calibration data appear in Tables 2 and 3. The refractive index data for acetic acid-propionic acid could be

Table 2. Refractive index at 25°C of mixtures of formic acid and propionic acid

x_A	h	x_A	n
0.028	1.3834	0.557	1.3770
0.094	1.3825	0.574	1.3766
0.133	1.3821	0.652	1.3756
0.154	1.3817	0.675	1.3751
0.240	1.3809	0.709	1.3746
0.255	1.3808	0.712	1.3747
0.279	1.3805	0.729	1.3745
0.281	1.3801	0.753	1.3740
0.294	1.3799	0.768	1.3735
0.382	1.3794	0.770	1.3735
0.391	1.3790	0.782	1.3734
0.421	1.3785	0.838	1.3722
0.473	1.3780	0.854	1.3720
0.490	1.3777	0.929	1.3706
0.502	1.3775	0.955	1.3701
0.535	1.3772	0.974	1.3695

represented by the straight line $n = -0.0148x_A + 1.3841$ so that the accuracy in estimating the concentration data is at least 0.02 concentration units. For formic acid-propionic acid $n = -0.0122x_A + 1.3837$ in the region where $0 < x_1 < 0.6$ and $n = -0.02x_A + 1.389$ for

Table 3. Refractive index at 25°C of mixtures of acetic acid and propionic acid

x_A	n	x_A	n
0.012	1.3840	0.317	1.3795
0.013	1.3837	0.345	1.3790
0.017	1.3839	0.385	1.3785
0.023	1.3836	0.429	1.3780
0.043	1.3835	0.438	1.3775
0.048	1.3835	0.559	1.3760
0.062	1.3831	0.594	1.3745
0.073	1.3830	0.752	1.3730
0.083	1.3826	0.860	1.3715
0.121	1.3820	0.885	1.3710
0.218	1.3810	0.906	1.3705
0.232	1.3805	0.986	1.3695

$X_A > 0.7$ so that the data accuracy is the same as for the first system.

RESULTS AND DISCUSSION

Isobaric measurements at 760 mm Hg of the vapor-liquid equilibrium were carried out for the associating solutions formic acid-propionic acid and acetic acid-propionic acid. The main results are summarized in Tables 4 and 5 and Figs. 1 to 3.

Three models were used to explain the experimental results: (1) The simplest one which assumes ideal behaviour in both phases and neglects association due to hydrogen bonding. (2) A more developed model which

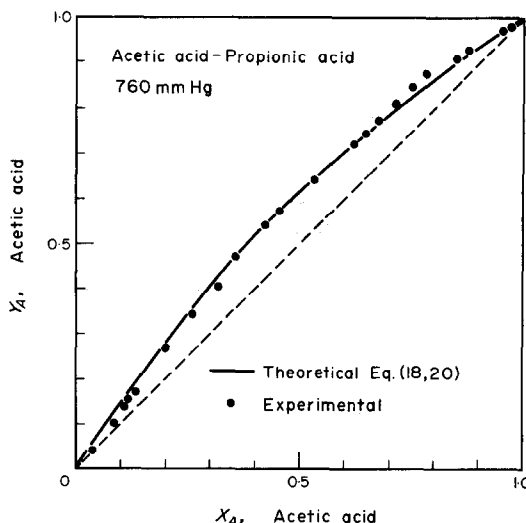


Fig. 1. Comparison between experimental and theoretical data for y_A and x_A .

accounts for mono and hetero-dimerization in the vapour phase, but assumes ideal solution behaviour in both phases. (3) A rather general model which accounts for association in both phases as well as non-ideal solution behaviour in the liquid phase.

Model 3 is the one that has been described in the previous paragraphs and has been applied to the

Table 4. Vapour-liquid equilibrium data for acetic acid-propionic acid at 760 mm Hg

$t^\circ\text{C}$	Experimental		Eq. (11) γ_A	Eq. (14) γ_B	Eq. (12) Z_A	Eq. (15) Z_B	Theoretical (Eq. 18, 20)	
	x_A	y_A					x_A	y_A
138.95	0.035	0.040	0.7345	1.0371	0.5613	0.5843	0.068	0.106
137.35	0.085	0.105	0.8223	1.0563	0.5495	0.5719	0.128	0.193
137.10	0.105	0.140	0.8925	1.0434	0.5477	0.5700	0.138	0.206
136.55	0.115	0.155	0.9131	1.0494	0.5436	0.5657	0.159	0.235
135.85	0.135	0.175	0.8917	1.0646	0.5384	0.5602	0.186	0.271
134.35	0.200	0.275	0.9772	1.0454	0.5272	0.5485	0.246	0.346
132.25	0.260	0.345	0.9867	1.0692	0.5116	0.5320	0.333	0.447
130.95	0.320	0.410	0.9797	1.0783	0.5019	0.5217	0.388	0.507
130.25	0.360	0.475	1.0241	1.0352	0.4967	0.5162	0.418	0.539
129.25	0.420	0.545	1.0288	1.0119	0.4892	0.5083	0.462	0.583
128.75	0.445	0.580	1.0444	0.9868	0.4855	0.5043	0.484	0.604
126.55	0.535	0.650	1.0199	1.0298	0.4691	0.4870	0.584	0.696
125.15	0.620	0.720	1.0039	1.0393	0.4588	0.4759	0.649	0.751
124.15	0.650	0.750	1.0185	1.0297	0.4514	0.4681	0.697	0.790
124.05	0.675	0.780	1.0221	0.9779	0.4506	0.4673	0.702	0.793
124.40	0.715	0.815	1.0009	0.9307	0.4532	0.4700	0.685	0.780
122.35	0.750	0.850	1.0385	0.8995	0.4381	0.4539	0.784	0.856
121.25	0.800	0.880	1.0311	0.9212	0.4300	0.4453	0.839	0.895
120.70	0.850	0.915	1.0206	0.8805	0.4260	0.4410	0.866	0.914
120.05	0.855	0.915	1.0283	0.9238	0.4213	0.4360	0.899	0.936
120.20	0.875	0.930	1.0181	0.8797	0.4224	0.4372	0.892	0.931
118.50	0.950	0.975	1.0179	0.8150	0.4100	0.4240	0.979	0.987
118.05	0.970	0.985	1.0165	0.8230	0.4068	0.4205	1.002	1.001

Table 5. Vapour-liquid equilibrium data for formic acid-propionic acid

Experimental			Theoretical (Eqs. 18, 20).					
$t/^{\circ}\text{C}$	x_A	y_A	Eq. (11) γ_A	Eq. (14) γ_A	Eq. (12) Z_A	Eq. (15) Z_B	x_A	y_A
135.55	0.025	0.100	1.9681	1.0374	0.7922	0.5580	0.098	0.198
134.90	0.050	0.150	1.4938	1.0203	0.7865	0.5531	0.111	0.221
134.50	0.070	0.210	1.5031	0.9777	0.7822	0.5502	0.119	0.234
131.15	0.110	0.255	1.2457	1.0373	0.7592	0.5240	0.187	0.344
129.05	0.150	0.310	1.1591	1.0539	0.7435	0.5078	0.231	0.408
126.65	0.210	0.395	1.1072	1.0490	0.7249	0.4894	0.284	0.477
124.45	0.240	0.440	1.1291	1.0596	0.7082	0.4723	0.334	0.537
123.25	0.265	0.465	1.1076	1.0749	0.6990	0.4631	0.362	0.568
120.75	0.300	0.505	1.1188	1.1033	0.6796	0.4437	0.422	0.629
118.55	0.375	0.580	1.0746	1.1014	0.6618	0.4272	0.477	0.680
116.05	0.420	0.625	1.0882	1.1197	0.6417	0.4082	0.541	0.735
115.15	0.450	0.655	1.0840	1.1085	0.6343	0.4016	0.565	0.753
112.85	0.500	0.690	1.0771	1.1523	0.6156	0.3843	0.628	0.799
111.59	0.575	0.725	1.0096	1.2369	0.6051	0.3751	0.663	0.823
109.53	0.620	0.755	1.0167	1.2894	0.5881	0.3600	0.722	0.860
105.72	0.750	0.850	1.0217	1.3060	0.5561	0.3330	0.835	0.924
104.43	0.820	0.900	1.0155	1.2454	0.5452	0.3242	0.875	0.944
103.05	0.875	0.925	1.0056	1.3866	0.5336	0.3146	0.919	0.964
101.71	0.955	0.970	0.9924	1.5880	0.5223	0.3057	0.962	0.984

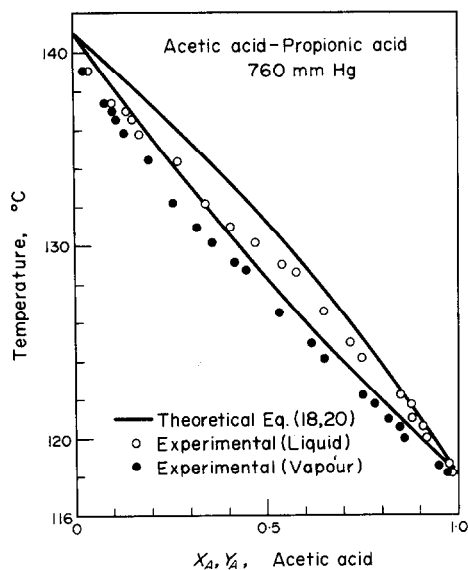


Fig. 2. Comparison between experimental and theoretical data for the temperature.

experimental data together with the following physical properties:

Vapour pressure of formic acid[10]

$$\log P_A^* = 7.3779 - \frac{1563.28}{t + 247.06} \quad (21)$$

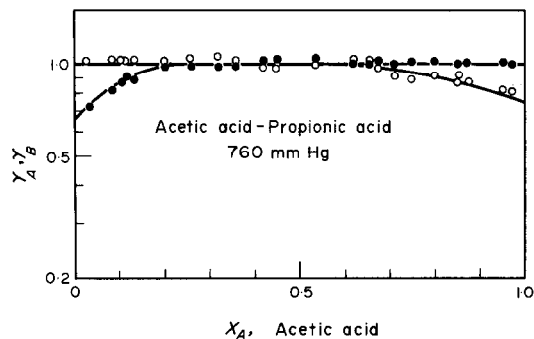


Fig. 3. Overall activity coefficients.

Vapour pressure of acetic acid[10]

$$\log P_A^* = 14.39756 - \frac{9399.86}{t + 698.0} \quad (22)$$

Vapour pressure of propionic acid[10]

$$\log P_B^* = 7.5476 - \frac{1617.06}{t + 205.67} \quad (23)$$

The dimerization constants for the vapour are:

For formic acid[9]

$$\log K_{A_2} = -10.743 + \frac{3083}{T} \quad (24)$$

For acetic acid[10]

$$\log K_{A_2} = -10.4205 + \frac{3166}{T} \quad (25)$$

For propionic acid[9]

$$\log K_{B_2} = -10.843 + \frac{3316}{T} \quad (26)$$

For the hetero-dimerization constant, we use the relation[11, 16]

$$K_{AB} = 2\sqrt{K_{A_2}K_{B_2}} \quad (27)$$

For the system acetic acid-propionic acid, the simplest model was found to be inconsistent with the Gibbs-Duhem relation hence the experimental data were analyzed using Model 2 which assumes $\gamma_A = \gamma_B = 1$. Figures 1 and 2 show the equilibrium data calculated by means of Eqs. (18)–(20) using the measured values of the total pressure (760 mm Hg) and temperatures. As noted from Fig. 1 the agreement between the experimental results (circles) and the computed ones (solid line) is very good. In other words, Model 2 will give a good estimation for the $x-y$ relationship. The calculated boiling point curve is shown in Fig. 2 by the solid line and indicates that temperature predictions will be higher by about 2 per cent than the experimental results. It should be noted that the agreement becomes better for larger values of X_A . The fitting can be improved by dropping the condition that the activity coefficients are unity and calculating their values by means of the simultaneous solution of Eqs. (16) and (17) and the appropriate physical properties as given by Eqs. (21)–(27)-Model 3. The corresponding values of γ_A and γ_B appear in Table 4. It may be seen that the overall activity coefficients calculated according to the new model are close to unity indicating that the solution acetic acid-propionic acid has only a slight negative deviation from ideality. Area tests of the values given in Table 4 and Fig. 3 also indicate that the activity coefficients are thermodynamically consistent. The above behaviour is in close agreement with the isothermal results at 20°C measured by Christian[11]. It was also found that the values of γ_A and γ_B becomes inconsistent when heterodimerization in the vapour was neglected. Table 4 also contains values of the associated parameters Z_A , Z_B that permit calculating the true mole fractions of the monomer, dimer and heterodimer.

The results for the system formic acid propionic acid are given in Table 5. In this case, the agreement between the theoretical model and the experimental data is not as good as before. For a given x_A , y_A data, the maximum deviation between the measured and calculated tempera-

ture was around 5 per cent. Application of the non-ideal model yielded the values of γ_A , γ_B detailed in Table 5. However, the area test revealed a deviation of about 25 per cent which increased to about 200 per cent when hetero-dimerization was omitted from Model 3 ($K_{AB} = 0$ in Eqs. (16), (17)). Based on this result, the 25 per cent deviation may be attributed to the neglect of the fact that higher polymerizations in the mixture are also possible in addition to the mono and hetero-dimerization considered in our model. Efforts to extend the model to include higher polymerizations were not successful due to the lack of information about the respective association constants in the vapour phase.

The data for γ_A and γ_B appearing in Tables 4 and 5 were correlated by means of the ratio $\ln(\gamma_A/\gamma_B)$ applying the Simplex method. For Margules' equation we used the expressions

$$\ln \gamma_A = X_B^2(2B - A) + 2X_B^3(A - B) \quad (28)$$

$$\ln \gamma_B = X_A^2(2A - B) + 2X_A^3(B - A) \quad (29)$$

For Redlich-Kister the following expansions were found adequate[16]

$$\ln \gamma_A = (A + 3B + 5C)X_B^2 - 4(B + 4C)X_B^3 + 12CX_B^4 \quad (30)$$

$$\ln \gamma_B = (A - 3B + 5C)X_A^2 + 4(B - 4C)X_A^3 + 12CX_A^4 \quad (31)$$

Wilson's equations were correlated for isothermal and non-isothermal energy parameters as follows[16, 17]

$$\ln \gamma_A = -\ln(X_A + \Lambda_{12}X_B) + X_B \left[\frac{\Lambda_{12}}{X_A + \Lambda_{12}X_B} - \frac{\Lambda_{21}}{\Lambda_{21}X_A + X_B} \right] \quad (32)$$

$$\ln \gamma_B = -\ln(X_B + \Lambda_{21}X_A) - X_A \left[\frac{\Lambda_{12}}{X_A + \Lambda_{12}X_B} - \frac{\Lambda_{21}}{\Lambda_{21}X_A + X_B} \right] \quad (33)$$

where

$$\Lambda_{12} = \frac{v_B}{v_A} \exp \left[-\frac{\lambda_{12} - \lambda_{11}}{RT} \right] \quad (34)$$

$$\Lambda_{21} = \frac{v_A}{v_B} \exp \left[-\frac{\lambda_{21} - \lambda_{22}}{RT} \right] \quad (35)$$

and

$$\lambda_{12} - \lambda_{11} = A_1 + A_2(T - 273.15) \quad (36)$$

$$\lambda_{21} - \lambda_{22} = B_1 + B_2(T - 273.15) \quad (37)$$

v_A, v_B are the molar volumes of the pure components. The corresponding parameters appear in Table 6, together with the values of the sum of squares of the deviations

Table 6. Parameters for the activity coefficients equations

	Acetic acid-Propionic acid	Formic acid-Propionic acid
Margules		
A	-0.2509	0.4969
B	-0.1876	0.2296
SSQ	0.0569	0.2463
Redlich Kister		
A	-0.1964	0.3169
B	0.0269	-0.0833
C	-0.1256	0.2411
SSQ	0.0327	0.1613
Wilson		
1. Isothermal		
Λ_{12}	0.5494	0.2659
Λ_{21}	1.8766	1.6832
SSQ	0.0561	0.1975
2. Non isothermal		
A_1	1.7485	-0.2698
A_2	1.6258	0.7239
B_1	1.4975	2.1006
B_2	-2.0348	2.6971
SSQ	0.0522	0.2728

(SSQ). It should be noted that correlation of the data by means of the ratio $\ln(\gamma_A/\gamma_B)$ gave better results with respect to the difference

$$\frac{\gamma_{A,\text{expt.}} - \gamma_{A,\text{calc.}}}{\gamma_{A,\text{expt.}}} 100$$

as compared to a correlation based on $G^E = X_A \ln \gamma_A + X_B \ln \gamma_B$. The maximum deviation was around 10 per cent. and the average 5 per cent.

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NOTATION

- A species A (monomer + dimer); parameters in Eqs. (28)–(31)
 A_1 monomer of A ; parameter in Eq. (36)
 A_2 dimer of A ; parameter in Eq. (36)
 B species B (monomer + dimer); parameters in Eqs. (28)–(31)
 B_1 monomer of B ; parameter in Eq. (37)
 B_2 dimer of B ; parameter in Eq. (37)
 AB heterodimer formed by $A_1 + B_1$
 G^E excess Gibbs energy

- K_{AA} vapour phase equilibrium constant for the formation of A_2 , $\text{mm}^{-1} \text{Hg}$
 K_{BB} vapour phase equilibrium constant for the formation of B_2 , $\text{mm}^{-1} \text{Hg}$
 K_{AB} vapour phase equilibrium constant for the formation of AB , $\text{mm}^{-1} \text{Hg}$
 n number of moles; refractive index
 P total pressure, mm Hg
 P_i° vapour pressure of the pure species i ($i = A, B$), mm Hg
 $P_{i_1}^\circ$ vapour pressure of the pure monomer of species i ($i = A, B$), mm Hg
 $P_{i_2}^\circ$ vapour pressure of the pure dimer of species i ($i = A, B$), mm Hg
 R gas constant
 t, T temperature, $^\circ\text{C}$, $^\circ\text{K}$
 v_i molar volume of pure species i , cm^3/mol ($i = A, B$)
 x_i, y_i stoichiometric mole fraction of species i in the liquid phase; in the vapour phase, ($i = A, B$)
 Z factor expressing influence of association in the vapour phase, defined by Eqs. (12), (15)
 γ_A, γ_B activity coefficient for the monomer of A (or B) in the mixture
 γ_A, γ_B overall activity coefficients defined by Eqs. (12) and (15)
 ξ_i, η_i true mole fractions of species i in the liquid phase; in the vapour phase ($i = A, B$)
 λ_{ij} energy of interaction
 Λ_{ij} Wilson parameter

Subscripts and superscripts

- A the pure and the more volatile species; formic or acetic acids
 B the pure and the less volatile species; propionic acid
 $1, 2$ monomer, dimer
 0 pure species

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